# "What Were They Thinking?" Using Open-Text Responses to Validate Constructs in Survey Experiments

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**Abstract**

Survey experiments enable causal inference, but researchers rarely verify whether respondents process experimental stimuli as intended. This paper introduces a practical framework for construct validation using open-text responses embedded in survey experiments, demonstrating two complementary approaches: structural topic modeling (STM) to discover what respondents discussed, and large language model (LLM) classification to confirm whether responses align with theoretically expected constructs. Applying this framework to conjoint experiments on immigrant naturalization in South Korea ($N = 1{,}999$) and Taiwan ($N = 2{,}050$), I find that a social integration cue shifts respondent reasoning, in both countries, from economic and practical criteria toward civic engagement themes. Two independent methods converge on this finding: data-driven STM identifies civic topics that increase 7–11 percentage points when the cue is shown, while theory-driven LLM classification shows civic commitment rising 15–18 percentage points. Six independent LLM classifiers (Fleiss' $\kappa = 0.745$), spanning 3B to 72B parameters, produce consistent results. A second application to a standard conjoint on immigrant admission (without within-design treatment randomization) demonstrates that the framework transfers to the more common experimental setup, and validation against expert human-coded ground truth from a third conjoint confirms that iterative codebook refinement produces substantial LLM-human agreement. The paper offers practical guidance on iterative codebook design informed by topic modeling, multi-model cross-validation for reproducibility, and accessible infrastructure using open-weight models on consumer hardware.

**Keywords:** survey experiments, construct validation, open-ended questions, structural topic models, large language models, text classification, immigration attitudes

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# Introduction

Survey experiments are among the most common designs in contemporary social science. Random assignment to treatment conditions allows researchers to estimate causal effects on attitudes, preferences, and behaviors. Yet an assumption underlying these designs often goes untested: that respondents process experimental stimuli as intended and that closed-ended response items capture the constructs researchers claim to measure.

This assumption is especially tenuous in conjoint experiments (Hainmueller et al., 2014), where respondents evaluate multi-attribute profiles. A conjoint profile may contain ten or more randomly varied attributes, but researchers typically interpret treatment effects as if respondents attended to and processed each attribute independently. The possibility that respondents focused on entirely different aspects of the profile, or interpreted attributes in unexpected ways, is rarely examined.

I propose a straightforward solution: embed open-text questions in survey experiments and use computational text analysis to systematically examine respondent reasoning. When a respondent is asked "Why did you choose this person?" after a conjoint task, the answer shows the reasoning behind the observed choice. If the experimental manipulation shaped respondent reasoning as intended, this should be visible in the open-text responses.

I develop and demonstrate a two-pronged analytical framework. The first approach is structural topic modeling (STM) for *discovery*: what topics did respondents discuss, and how did experimental conditions affect topic prevalence? STM is well-suited to this task because it incorporates treatment conditions as covariates during model estimation (Roberts et al., 2014). The second is large language model (LLM) classification for *confirmation*: do open-text responses align with theory-derived categories? Following recent advances in codebook design for LLM measurement (Halterman and Keith, 2025), I classify responses into substantive codes that correspond to the constructs the experimental design aims to activate.

The empirical application uses original conjoint survey data on attitudes toward immigrant naturalization in South Korea ($N = 1{,}999$) and Taiwan ($N = 2{,}050$), collected in 2024, extending conjoint research on immigration attitudes (Hainmueller and Hopkins, 2015; Denney and Green, 2021). The con-

joint design includes a "social integration" attribute that randomly describes immigrants' civic engagement activities. The validation question is direct: when respondents see the civic engagement cue, do they actually write about civic engagement?

They do. In both countries, the social integration cue increases discussion of civic and community themes while reducing discussion of economic and credential-based criteria. These effects are substantively large (7–18 percentage points depending on the method and country) and consistent across both analytical approaches and six independent LLM classifiers (Fleiss' $\kappa = 0.745$), supporting the construct validity of the "cues of commitment" framework underlying the conjoint design. To test portability beyond this intervention-based design, I apply the same framework to a standard conjoint on immigrant admission (Denney and Green, 2021), where no within-design treatment randomization is available. LDA and LLM classification converge on the same constructs, showing that the approach works for both intervention-based and conventional conjoint designs.

The contribution is primarily methodological. I develop a two-stage workflow for construct validation: unsupervised topic modeling first discovers the empirical structure of open-text responses, and those findings then inform the design of a theory-driven codebook for LLM classification. Unlike approaches that apply LLM classification or scaling directly to text (Mellon et al., 2024; DiGiuseppe and Flynn, 2026), this workflow uses unsupervised findings to check and refine the codebook before classification begins. In this case, the initial codebook conflated two constructs that the STM showed respond differently to the experimental treatment; the revised version separated them, and the resulting classification was sharper for it. I apply the same workflow to two further datasets: a standard conjoint on immigrant admission (without within-design treatment randomization) and a conjoint with expert human-coded ground truth, where iterative codebook refinement brings LLM-human agreement to $\kappa \approx 0.6$–$0.7$. All classification is done in six models—one proprietary (GPT-4o-mini) and five open-weight, from 3B to 72B parameters—to test reproducibility across model families and sizes (Barrie et al., 2025; Chae and Davidson, 2025). The two smallest models run on a consumer GPU.

# Background and Related Work

## Open-Ended Questions in Survey Research

Open-ended questions have a long history in survey research, but their systematic use for construct validation is less established. Geer (1988) showed that open-ended responses capture genuine attitudes rather than mere articulateness. Lupia (2018) documented persistent quality problems in coding open-ended responses and called for greater transparency. Glazier et al. (2021) introduced a "self-coding" method where respondents categorize their own open-ended responses, a direct check on whether researcher-imposed categories match respondent interpretations. Ziegler (2022) uses text-as-data methods on open-ended survey responses as manipulation checks. Haaland et al. (2025) review the growing use of open-ended survey data to uncover mechanisms behind beliefs and behavior, including LLM-based annotation at scale.

This paper extends that line of work by using computational text analysis to scale from small hand-coded samples to entire survey populations.

## Structural Topic Models

Topic models provide a data-driven approach to discovering the thematic structure of text corpora (see Grimmer and Stewart, 2013; Grimmer et al., 2022; Gentzkow et al., 2019).[1] Roberts et al. (2014) developed the Structural Topic Model for contexts where document-level metadata (such as experimental treatment conditions) should inform topic estimation; Roberts et al. (2016) provide the formal statistical foundation. Unlike standard LDA, STM allows covariates to affect both topic prevalence (which topics appear in which documents) and topic content (which words are associated with which topics). This makes STM ideal for survey experiments: researchers can directly estimate how experimental conditions affect the topics respondents discuss (see also Egami et al., 2022; Fong and Grimmer, 2023, on causal inference with text).

[1] For the foundational Latent Dirichlet Allocation model, see Blei et al. (2003).

Hobbs and Green (2025) distinguish between topic discovery and attitude inference, arguing that topic models excel at the former but require careful interpretation for the latter. The present application is primarily concerned with topic discovery (identifying what respondents discussed) and with treatment effects on topic prevalence, so STM is a natural fit.

## LLM-Based Text Classification

Large language models are now widely used for text classification (Gilardi et al., 2023; Ziems et al., 2024; Bail, 2024). Chae and Davidson (2025) compare zero-shot, few-shot, and instruction-tuned LLMs for text classification in social science, finding that model size and prompting strategy jointly determine classification quality. The present study tests this directly by deploying models from 3B to 72B parameters. Tornberg (2025) demonstrates that LLMs can outperform human coders on contextual reasoning tasks. Heseltine and Clemm von Hohenberg (2024) show that hybrid human-LLM workflows achieve 93%+ accuracy cost-effectively. Ornstein et al. (2025) provide the `prompt` R package for few-shot LLM classification. Mellon et al. (2024) demonstrate that LLMs can code open-text survey responses at scale with accuracy comparable to human coders, while Rathje et al. (2024) show that GPT performs well for text analysis across 12 languages, supporting multilingual classification. Most of this work assigns each response to a discrete code. DiGiuseppe and Flynn (2026) take a different tack, using LLM pairwise comparisons and Bradley-Terry modeling to place responses on a continuous scale. The choice between categorical and continuous measurement depends on the research question; the present study asks what *kind* of reasoning respondents use, so categorical codes are the natural fit.

Codebook design matters more than model choice. Halterman and Keith (2025) argue that the quality of LLM classification depends on the precision of the codebook: loose instructions lead to poor performance regardless of model capability. They recommend codebooks with explicit definitions, inclusion criteria, exclusion criteria, and worked examples.

Reproducibility is an equally pressing concern. Barrie et al. (2025) show that proprietary LLM APIs exhibit substantial variance across runs and model versions, threatening the reproducibility of research findings. They recommend open-weight models that can be run locally with fixed parameters, producing

deterministic outputs (see also Le Mens and Gallego, 2025). This concern motivates the present study's design: five open-weight models form the reproducible core, while GPT-4o-mini is included as a reference point for the proprietary models that dominate current practice in political science. Gilardi et al. (2023) used GPT-3.5-turbo, Tornberg (2025) and Heseltine and Clemm von Hohenberg (2024) used GPT-4, and Chae and Davidson (2025) used GPT-4o. The open-weight results show that researchers need not rely on proprietary APIs to achieve comparable classification quality.

# Data and Research Design

## The Conjoint Experiment

The data come from original surveys conducted in South Korea and Taiwan between January and June 2024, examining public attitudes toward immigrant naturalization (Denney et al., 2025). Respondents were recruited via Qualtrics using stratified quota sampling by sex, age, and geographic region to approximate national census distributions. The final samples comprise $N = 1{,}999$ in South Korea (51% male) and $N = 2{,}050$ in Taiwan (53% male).

Each respondent evaluated paired immigrant profiles in a forced-choice conjoint experiment (Hainmueller et al., 2014; Bansak et al., 2021). In South Korea, respondents completed 10 paired-choice tasks; in Taiwan, 7 tasks. Profiles varied on 12 randomly assigned attributes covering demographic characteristics, economic qualifications, and social integration (see Supplementary Information, Table A.2 for the full attribute table).

The attribute of interest for this analysis is *social integration*, which describes the immigrant's civic engagement activities with four levels: "only associates with fellow expatriates," "often spends time alone," "co-hosts monthly multicultural events," and "volunteers weekly at a local community center." Due to the randomization design, this attribute was randomly withheld from some profiles, creating a clean cue-shown vs. cue-not-shown comparison.

After the conjoint task, respondents answered an open-text question: *"Why did you choose this person as more deserving of citizenship?"* The question appeared once, after the final conjoint choice, and was

required (respondents could not skip it). Responses ranged from single words to several sentences. No minimum length was imposed, and no responses were excluded on the basis of brevity alone.

### Open-Text Corpus

The corpus comprises 4,049 responses: 1,999 from South Korea (Korean) and 2,050 from Taiwan (Traditional Chinese). Responses are brief (median: 10 characters in Korean, 6 in Chinese) but substantive. Approximately 70% of responses contain at least one identifiable reference to a conjoint attribute.

The treatment variable is `cue_shown`: whether the social integration attribute was displayed on the conjoint profile the respondent evaluated. Due to the attribute randomization design, approximately half of responses in each country come from profiles where the social integration cue was shown (South Korea: 49.8%; Taiwan: 49.1%).

## Analytical Framework

### Approach 1: Structural Topic Modeling

I fit separate STM models for each country, using the `stm` package in R (Roberts et al., 2019) with spectral initialization and seed = 42. Korean text is preprocessed using the `kiwipiepy` morphological analyzer and Chinese text using `quanteda`'s built-in segmenter (Benoit et al., 2018). Preprocessing follows established guidelines for multilingual text analysis (Lucas et al., 2015; Denny and Spirling, 2018).

Since the conjoint design is fully randomized, the prevalence specification includes only the treatment indicator:

$$\text{prevalence} \sim \texttt{cue_shown} \tag{1}$$

I select the number of topics ($K$) using the coherence-exclusivity tradeoff from `searchK()` diagnostics, resulting in $K = 6$ for South Korea and $K = 8$ for Taiwan.

Treatment effects on topic prevalence are estimated using `estimateEffect()`, which provides coefficient estimates, standard errors, and confidence intervals for the effect of `cue_shown` on each topic's

prevalence.

## Approach 2: LLM Classification

I developed the codebook iteratively, using STM findings to guide refinement. The initial version (v1) used four codes: `civic_commitment`, `identity_concern`, `economic_focus`, and `other`. The civic commitment code captured civic engagement, volunteering, and community involvement, but also included language competence and length of residence as signals of integration commitment.

Running v1 across GPT-4o-mini exposed a problem: 561 responses (13.9%) were coded as civic commitment based solely on language fluency or residence duration, which are practical attributes, not active civic engagement. The STM results showed that topics about language and residence *decrease* when the civic integration cue is shown, meaning these attributes behave as substitutes for civic engagement, not signals of it. Including them under civic commitment conflated participation in community life with individual-level adaptation to the host country.

This STM-informed diagnosis led to a revised codebook (v2) with five codes:

| Code | Definition |
|---|---|
| `civic_commitment` | Participation in community life: volunteering, community organizing, social participation. Outward-facing, prosocial engagement |
| `functional_integration` | Individual-level adaptation: language competence, residence duration, law-abiding behavior. Practical prerequisites for functioning in the host country |
| `identity_concern` | Ethnic, cultural, or national identity concerns, including religion and ethnicity when used as evaluation criteria |
| `economic_focus` | Economic contributions: jobs, taxes, career, occupation, work experience |
| `other` | Truly uninformative responses: filler words, gibberish, meta-comments. Brevity alone is never a reason for this code |

**Table 1:** LLM classification codebook (v2). Codes follow the design principles of Halterman and Keith (2025), with explicit definitions, inclusion/exclusion criteria, and few-shot examples in both Korean and Traditional Chinese.

The classification pipeline uses six LLMs. GPT-4o-mini (OpenAI API) provides a reference point

for the proprietary models used in prior work. Five open-weight models form the reproducible core:

| Model | Parameters | Infrastructure | Role |
|---|---|---|---|
| GPT-4o-mini | — | OpenAI API | Proprietary baseline; representative of models used in current practice |
| Qwen 2.5 3B | 3B | Consumer GPU | Open-weight; smallest model |
| Llama 3.1 8B | 8B | Consumer GPU | Open-weight; mid-range |
| Gemma 3 12B | 12B | University HPC (A100) | Open-weight; mid-range |
| Qwen 2.5 32B | 32B | University HPC (A100) | Open-weight; large |
| Qwen 2.5 72B | 72B | University HPC (A100) | Open-weight; largest model; size scaling |

**Table 2:** LLM classifiers used in the analysis. Open-weight models are run via the Ollama framework with temperature = 0 and seed = 42 for deterministic output.

All open-weight models are run with temperature = 0, seed = 42, and max_tokens = 200, producing deterministic outputs. Following Barrie et al. (2025), the open-weight models provide a reproducibility guarantee that does not depend on proprietary API stability. The range of model sizes (3B to 72B parameters, including three from the Qwen 2.5 family) allows a direct test of whether classification quality scales with model size (Chae and Davidson, 2025).

The classification pipeline is implemented in R and Python. Codebook generation (from a structured YAML definition following Halterman and Keith 2025), cross-model validation, and all statistical analyses are written in R; the OpenAI API classifier is a standalone Python script. Open-weight models are deployed through Ollama, a lightweight framework for running LLMs locally with fixed inference parameters. The two smallest models (3B, 8B) run on a consumer GPU (NVIDIA RTX 3060 Ti, 8GB VRAM); the three larger models (12B, 32B, 72B) run on a university high-performance computing cluster equipped with NVIDIA A100 GPUs, with the 32B and 72B models quantized to 4-bit precision to fit within GPU memory. All code, codebooks, and classification outputs are available in the replication archive.

Each response is classified independently by all six models. Because no human-coded ground truth exists for these responses, the analysis relies on cross-model consensus rather than any single model's judg-

ment. The majority-vote classification assigns each response the code that four or more of the six models agree on; responses without a four-model consensus (7.2% of the corpus) are excluded. This approach treats the six LLMs as independent raters, analogous to a team of human coders, and applies the same inter-rater reliability standards that are conventional in content analysis (Benoit et al., 2026). Majority-vote aggregation reduces the influence of model-specific biases (for example, Llama 3.1 8B's inflated functional integration rate in Taiwan) by requiring broad agreement across model families and sizes before any classification is assigned. Treatment effects are then estimated as differences in code prevalence between cue-shown and cue-not-shown conditions on this consensus-classified corpus, with bootstrap confidence intervals (2,000 iterations, percentile method) providing inference.

The individual models produce consistent results even without aggregation. Table C.6 in the Supplementary Information reports the cue effect for each model separately: all six agree on the direction of every substantive effect, with civic commitment increasing between +9.3 and +18.1 percentage points across all models and countries. Even the smallest model (Qwen 2.5 3B, 3 billion parameters, runnable on a consumer GPU) produces treatment effect estimates in the correct direction for every code in every country. The majority-vote procedure is thus a conservative safeguard, not a correction for disagreement.

## Triangulation

The two approaches answer different questions and, importantly, each one compensates for limitations in the other. STM shows the empirical structure of the text without imposing prior categories. It can surface patterns the researcher did not anticipate. But STM cannot tell you whether a topic corresponds to a specific theoretical construct, and on very short text (median 6–10 characters), topic assignments are often diffuse. LLM classification maps responses onto theory-derived categories with greater precision, but only if the categories are right. An ill-specified codebook produces precise but misleading estimates.

Running both methods on the same data creates a productive tension. The STM results in this study showed that language and residence topics move in the *opposite* direction from civic engagement topics under the treatment. That empirical pattern, which emerged without any prior categorization, made clear that the v1 codebook was grouping together constructs that respond differently to the experimental

manipulation. LLM classification alone would not have flagged this problem, because the model faithfully follows whatever codebook it is given. The STM provided the evidence needed to refine the measurement before drawing conclusions from it.

This is not simply triangulation in the sense of running two methods and checking whether they agree. It is a workflow in which each method has a specific role: STM identifies the empirical structure, LLM classification applies theoretical categories informed by that structure, and the combination produces more credible construct validation than either method alone. The key difference from single-stage LLM classification is the feedback loop: the unsupervised step diagnoses problems in the codebook that the supervised step, on its own, would faithfully reproduce.

A structural difference between the two methods also explains why they produce different effect magnitudes. STM assigns every document a probability distribution over all topics, including uninformative responses. Noise gets distributed across the topic space rather than isolated, producing conservative treatment effect estimates that reflect the full corpus (Roberts et al., 2014). LLM classification includes an explicit `other` category that filters uninformative responses out of the substantive codes, so treatment effects are measured against a cleaner denominator. The full-sample STM estimates are not wrong; they reflect a different estimand. The Supplementary Information (Appendix B) provides a sensitivity analysis showing that as diffusely assigned STM documents are progressively excluded, the STM civic effect increases monotonically and converges toward the LLM estimate.

# Results

## STM: What Did Respondents Write About?

The South Korean model ($K = 6$) identifies six topics that map onto conjoint profile attributes and respondent evaluation criteria. The largest topics concern identity and practical credentials: religious/ethnic preferences (22.9%), residence and assimilation effort (21.3%), and community engagement and volunteering (18.8%). Employment and family stability (13.1%), professional experience (12.1%), and language/visa considerations (11.9%) account for the rest. Taiwan's model ($K = 8$) produces a compara-

ble structure, with language fluency and first impressions (18.6%), work experience and social integration (18.5%), and civic activities and ethnicity (14.6%) as the largest topics.

That respondents wrote about the attributes they saw in the immigrant profiles is itself a finding. It means they processed the experimental stimuli and evaluated candidates on the dimensions the conjoint design intended to test.

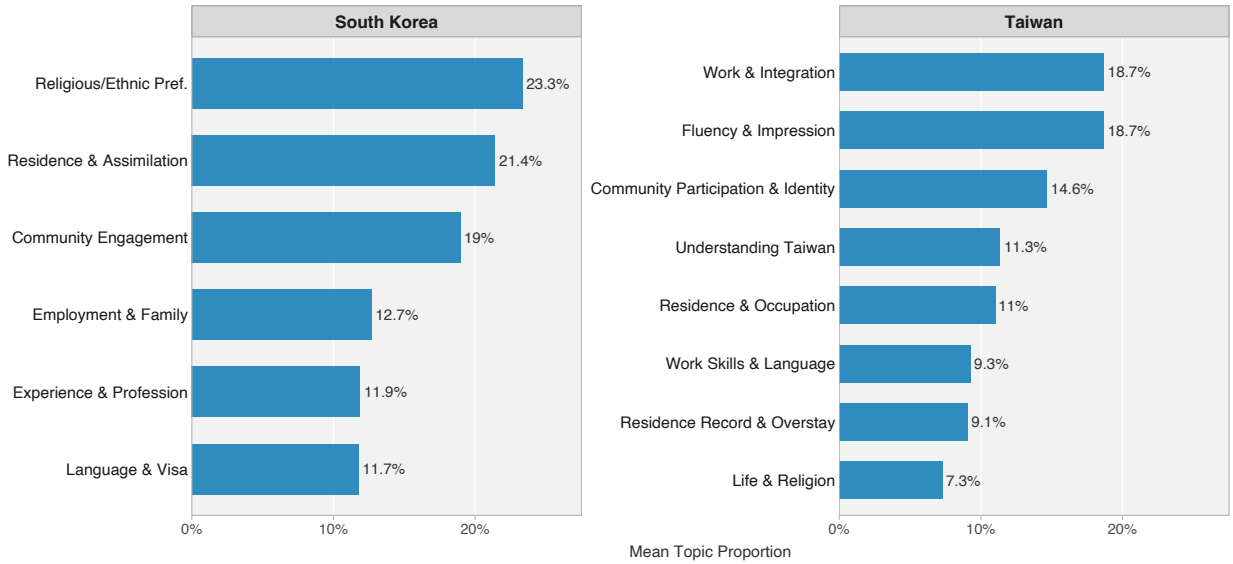


**Figure 1:** STM topic prevalence by country. Mean document-topic proportions for South Korea ($K = 6$) and Taiwan ($K = 8$).

## STM: Effect of the Social Integration Cue

In South Korea, the social integration cue increases discussion of Community Engagement by 6.7 percentage points while decreasing discussion of Language & Visa ($-4.0$ pp) and Employment & Family ($-2.3$ pp). All three effects are statistically significant.

Taiwan shows the same pattern at greater magnitude. The cue increases Civic Activities by 11.4 pp, Work & Integration by 3.1 pp, and Understanding Taiwan by 1.8 pp, while decreasing Fluency & Impression ($-5.4$ pp), Residence & Occupation ($-4.7$ pp), Residence Record ($-3.0$ pp), and Work Skills ($-2.7$ pp). All effects are significant at the 95% level; see Figure 2.

The pattern is consistent across both countries: when respondents see that an immigrant volunteers or participates in community life, they write about civic themes rather than practical credentials. The cue

does not just add a new topic to respondent reasoning; it shifts attention away from language, residence, and employment criteria.

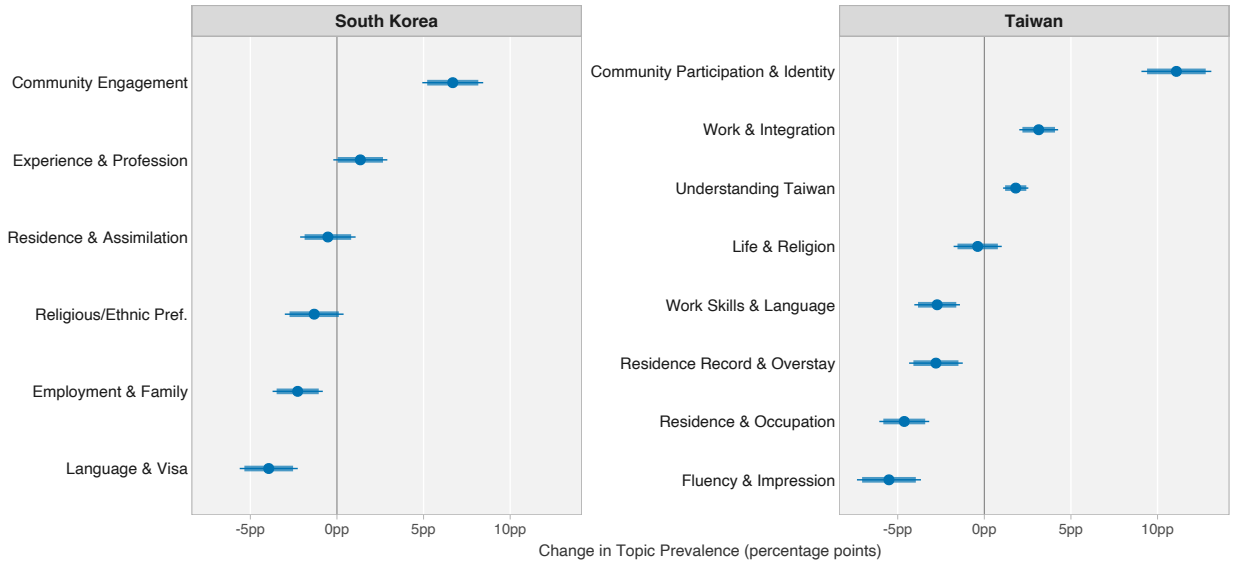


**Figure 2:** Effect of social integration cue on topic prevalence. STM coefficient estimates with 90% (thick) and 95% (thin) confidence intervals.

## LLM Classification: Code Prevalence

Using majority-vote classification (4+ of 6 models agree), functional integration is the most common substantive code in both countries (∼32%), followed by identity concern (15–21%), economic focus (19–23%), and other (20–21%). Civic commitment (active behavioral engagement) accounts for 2.0% (South Korea) and 1.0% (Taiwan) when the cue is *not* shown, rising to 16.6% and 19.2% when it *is* shown.

Separating functional integration from civic commitment makes clear that practical attributes (language, residence, law-abiding behavior) dominate respondent reasoning in both countries, distinct from the active civic engagement that the cue is designed to activate.

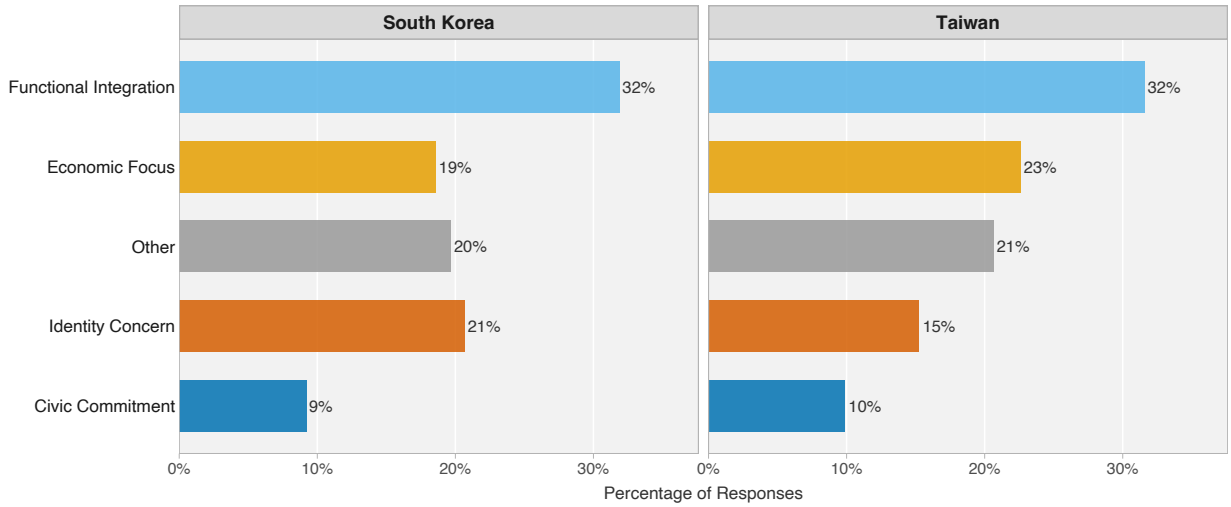


**Figure 3:** LLM classification: code prevalence by country. Majority-vote classification (4+ of 6 models agree).

## LLM Classification: Effect of the Social Integration Cue

In South Korea, the cue increases civic commitment by +14.7 percentage points (from 2.0% to 16.6%), while decreasing functional integration by $-7.9$ pp and economic focus by $-5.6$ pp. Identity concern decreases modestly ($-2.7$ pp).

The Taiwanese results are similar in direction but stronger in magnitude: civic commitment increases by +18.2 pp (from 1.0% to 19.2%), while economic focus decreases by $-11.4$ pp and functional integration by $-6.8$ pp. Identity concern is stable ($-1.0$ pp).

Bootstrap confidence intervals place all of these effects outside zero at the 95% level. Across the six individual models, civic commitment effects range from +9.3 to +16.1 pp in South Korea and +14.4 to +18.1 pp in Taiwan, all significant.

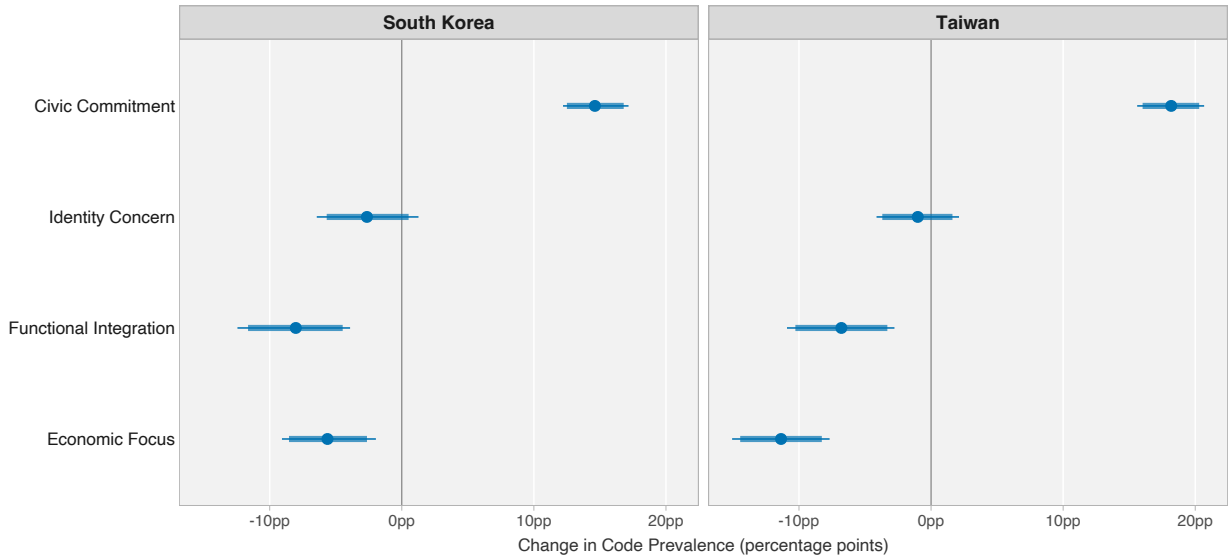


**Figure 4:** Effect of social integration cue on LLM-classified response themes. Majority-vote classification (4+ of 6 models). Bootstrap 90%/95% CIs (2,000 iterations).

Six independent LLMs produce consistent cue effect estimates (Figure 5; see also Table C.6 in the Supplementary Information for the full per-model breakdown). All six models agree that the civic integration cue significantly increases civic commitment and decreases economic focus in both countries. Pairwise agreement ranges from 66.3% to 91.2% (Cohen's $\kappa$: 0.578–0.886), with Fleiss' multi-rater $\kappa = 0.745$ indicating substantial agreement.

Agreement increases with model size. Within the Qwen 2.5 family, pairwise $\kappa$ with GPT-4o-mini rises monotonically: 0.744 (3B), 0.841 (32B), 0.860 (72B). The highest agreement overall is between Qwen 2.5 32B and 72B ($\kappa = 0.886$, 91.2% raw agreement), and the strongest cross-family pair is GPT-4o-mini and Qwen 2.5 72B ($\kappa = 0.860$, 89.1%). Larger open-weight models converge toward each other and toward the proprietary model. Chae and Davidson (2025) report a similar pattern.

Overall, 92.8% of responses receive a majority classification (4+ of 6 models agree). The weakest performer is Llama 3.1 8B, which inflates functional integration classifications in Taiwan (43.7% vs. 30–37% for other models), probably due to weaker CJK language capabilities. Yet even this model produces civic commitment effects in the correct direction (+12.2 pp in South Korea, +15.9 pp in Taiwan), and the smallest model in the study (Qwen 2.5 3B) agrees on the direction of every effect in every country. For short, semantically straightforward open-text responses, model size matters less for directional conclusions than

for precise agreement on individual classifications.

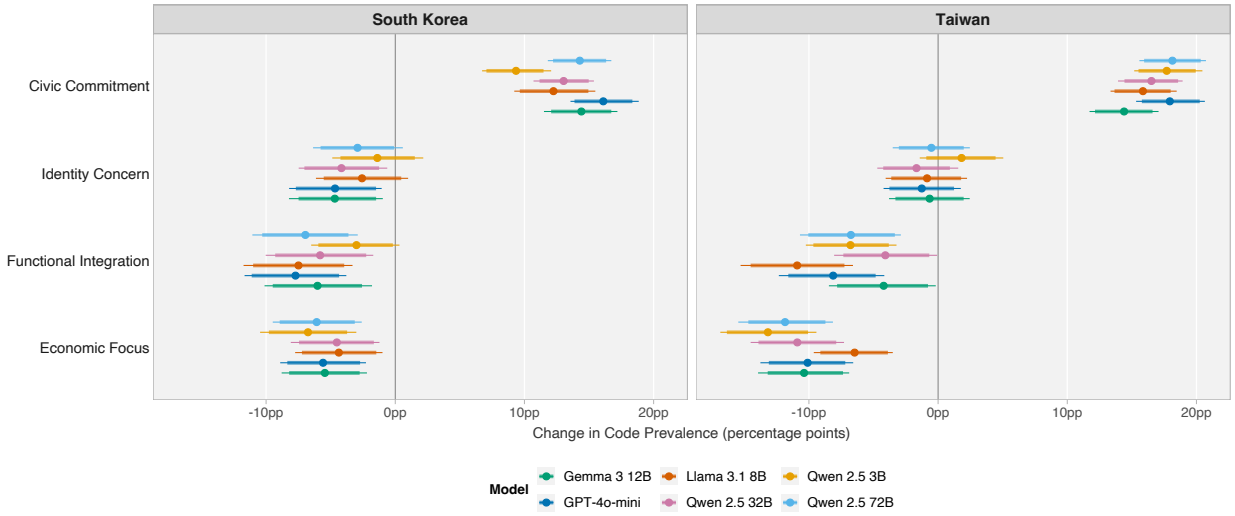


**Figure 5:** Cross-model robustness: cue effect estimates from six independent LLMs. All models produce consistent treatment effect estimates for civic commitment.

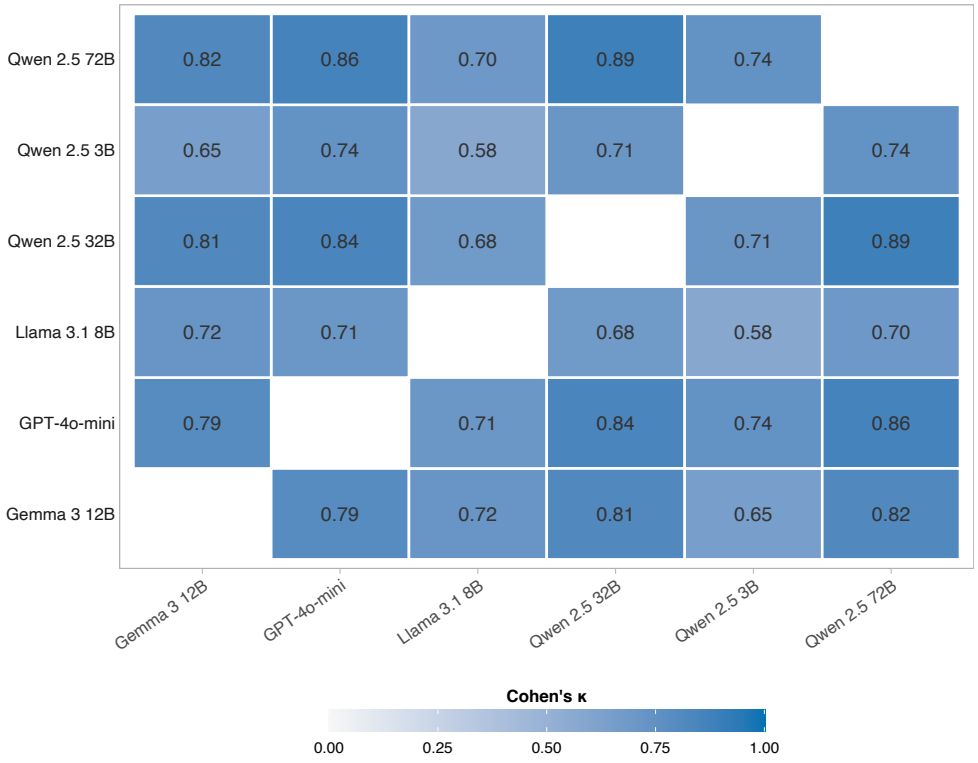


**Figure 6:** Cross-model agreement: pairwise Cohen's $\kappa$ for six LLMs. Fleiss' multi-rater $\kappa = 0.745$.

## Triangulation

The cleanest comparison between methods is for civic engagement, where a single STM topic maps directly to the LLM `civic_commitment` code in each country. In South Korea, the STM "Community Engagement" topic increases by 6.7 pp while the LLM civic code increases by 14.7 pp. In Taiwan, the STM "Community Participation & Identity" topic increases by 11.4 pp while the LLM civic code increases by 18.2 pp. All effects are significant. Both methods agree on the direction, significance, and cross-country pattern: the cue activates civic reasoning, and the effect is larger in Taiwan.

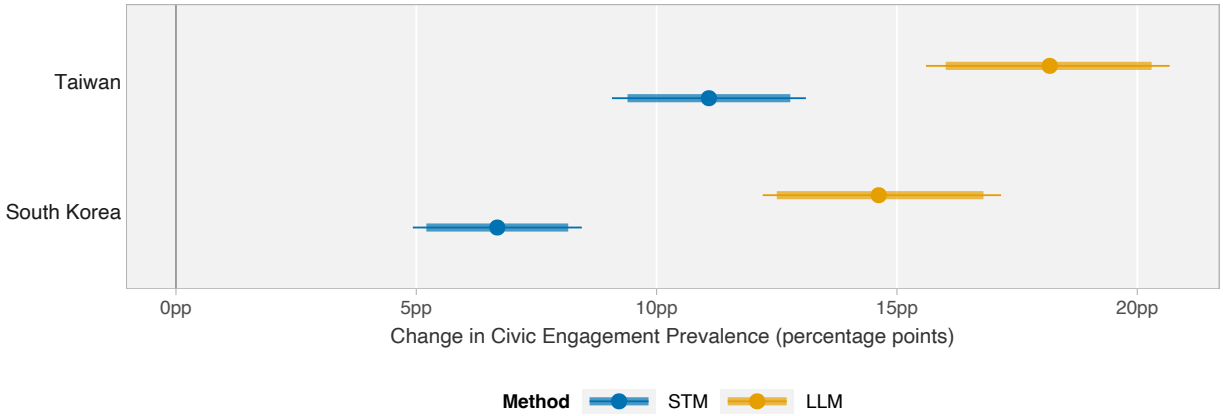


**Figure 7:** Triangulation: STM civic topic and LLM civic commitment treatment effects. The comparison where the topic-to-code mapping is unambiguous. Thick bars: 90% CIs; thin bars: 95% CIs.

Two factors explain why LLM estimates are larger than STM estimates. First, LLM codes are broader than individual STM topics: `civic_commitment` captures all civic reasoning in a single code, while STM distributes related content across multiple topics. Second, STM assigns topic proportions to every document regardless of content quality, while LLM classification routes uninformative responses to `other`, producing cleaner substantive codes and sharper treatment effects. This noise-dilution mechanism is quantifiable: the maximum topic probability ($\theta_{\max}$) for each document provides a measure of assignment clarity, and the median $\theta_{\max}$ is only 0.315 in South Korea and 0.303 in Taiwan (Supplementary Information, Appendix B). When diffusely assigned documents are progressively excluded, the STM civic effect rises monotonically and crosses the LLM estimate at $\theta_{\max} \approx 0.35$–$0.40$, which confirms that the gap between the two methods reflects how each handles uninformative text, not substantive disagree-

ment about the treatment effect.

For the remaining categories, the mapping between STM topics and LLM codes is not one-to-one, but the directional agreement is clear. On the practical and economic side, STM topics related to language, employment, and residence all decline when the cue is shown; the LLM similarly shows functional integration and economic focus declining. Identity concern is stable in both methods. Effects are consistently larger in Taiwan.

The v2 codebook refinement sharpens this convergence. By separating functional integration from civic commitment, the LLM classification now mirrors the STM finding that language and residence topics *decrease* when the cue is shown, which confirms that these practical attributes behave as substitutes for, not signals of, civic engagement.

## Extension: Framework Portability

The main analysis uses a conjoint design with within-design treatment randomization: the social integration attribute is randomly withheld, creating a clean cue-shown vs. cue-not-shown comparison that STM can incorporate as a prevalence covariate. Most conjoint experiments do not have this feature. To test whether the framework works in the more common setup, I apply it to a standard conjoint on immigrant *admission* to South Korea (Denney and Green, 2021), where all attributes are always shown and no within-design treatment randomization is available. I use a lighter setup: LDA topic modeling (replicating the original analysis) and LLM classification with two models (GPT-40-mini and Qwen 2.5 3B).

Denney and Green (2021) embedded open-text responses in a conjoint experiment examining South Korean attitudes toward immigrant admission ($N = 1{,}008$). Respondents evaluated paired immigrant profiles varying on seven attributes (reason for application, country of origin, language skills, profession, employment plans, gender, and ethnicity) and explained their choice in an open-text response.

The immigration attitudes literature identifies three competing explanations for how the public evaluates immigrants: sociotropic concerns (national-level norms and contribution), economic self-interest (labor market and fiscal concerns), and ethnocultural identity (origin-based preferences) (Hainmueller

and Hopkins, 2015). Applying the framework from the main analysis, I refine this trichotomy into five codes that distinguish functional integration (language ability, adaptation, settlement effort) from sociotropic reasoning. This parallels the civic commitment / functional integration distinction in the citizenship codebook and specifies what the original analysis left implicit.

| Code | Definition |
|---|---|
| Sociotropic | Social harmony, community benefit, public safety, normative fit |
| Economic | Employment, labor market, fiscal concerns (personal or national) |
| Integration | Language ability, cultural adaptation, settlement effort |
| Identity | Ethnicity, national origin, gender, religion, co-ethnic preference |
| Other | Uninformative, vague impressions, refusals |

**Table 3:** Classification codes for the immigration admission conjoint.

Turning to topic modeling, I replicate the original LDA analysis ($K = 6$) using Korean text tokenized with `kiwipiepy`. Table 4 shows the six topics with revised labels informed by the five-code framework. Two topics capture functional integration (language acquisition and Korean fluency/adaptation, together 36% of documents), one captures identity (co-ethnic preference, 17%), one captures sociotropic reasoning (societal benefit, 16%), one captures economic concerns (employment plans, 13%), and one is genuinely mixed (profession and nationality, 18%). The original paper's broad "sociotropic" label encompassed what are better understood as three distinct constructs: sociotropic, economic, and integration.

| Topic | Label | Framework | Top terms (English gloss) | % |
|---|---|---|---|---|
| 1 | General Profile Evaluation | Mixed | occupation, immigration, ethnicity, nationality | 18.0 |
| 2 | Employment Plans & Purpose | Economic | job-seeking, activity, asylum, plan | 13.4 |
| 3 | Co-ethnic Preference | Identity | co-ethnic, employment, North Korea, person | 16.8 |
| 4 | Integration Effort | Integration | language, woman, settlement, effort | 16.1 |
| 5 | Societal Benefit | Sociotropic | Korea, country, help, society | 15.8 |
| 6 | Language & Employment | Integration | Korean (lang.), fluent, adaptation, ability | 19.9 |

**Table 4:** LDA topics ($K = 6$) with revised framework labels.

Unlike the main analysis, I use LDA rather than STM because the original study did not include a treatment randomization amenable to prevalence covariates. LDA discovers document-level topic structure but cannot estimate treatment effects on topic prevalence directly; this limitation shows what STM adds when experimental metadata is available.

For LLM classification, GPT-4o-mini and Qwen 2.5 3B independently classify the 1,008 responses using the five-code codebook, achieving 79.2% agreement (Cohen's $\kappa = 0.737$). Cross-tabulating LDA topic assignments with LLM classifications shows convergence on the core constructs: for four of five non-mixed LDA topics, the dominant LLM code matches the expected framework label. Topic 2 (Employment) maps to economic, Topic 3 (Co-ethnic Preference) to identity, Topic 5 (Societal Benefit) to sociotropic, and Topic 6 (Language & Employment) to integration. The one exception is Topic 4 (Integration Effort), where many responses are too short for confident LLM classification and default to "other." Both models produce the same convergence pattern.

This extension addresses a practical question: does the framework require the specific features of the main analysis (within-design treatment randomization, six LLM classifiers, STM), or does it work under more typical conditions? The answer is that it transfers. LDA and LLM classification converge on the same constructs using a standard conjoint design, two classifiers, and a codebook adapted from the immigration attitudes literature.

The extension also illustrates the substantive value of the approach. What Denney and Green (2021) described broadly as "sociotropic" reasoning decomposes into three distinct constructs (sociotropic, economic, and integration), each with a corresponding LDA topic. Functional integration emerges as the single largest category in both LDA (36% of documents) and LLM classification (20–27%), yet it was underspecified in the original analysis. Applying the framework retrospectively to existing data can sharpen constructs that were too coarse in the original study.

One limitation is apparent: with only two models, model-specific biases are more visible. Qwen 2.5 3B systematically underclassifies short responses, defaulting to "other" where GPT-4o-mini identifies a substantive code. For applications with fewer models, the open-weight classifier provides a reproducibility check, but researchers should expect lower agreement on brief, ambiguous text.

# Extension: Validation against Human-Coded Ground Truth

The main analysis relies on cross-model consensus because no human-coded ground truth exists for the citizenship conjoint responses. A natural question is how well LLM classifications agree with independent human coding. To test this, I apply the framework to a dataset where expert human classifications are available: Denney et al. (2024), a conjoint experiment on South Korean attitudes toward North Korean migrant entrepreneurship policy ($N = 2{,}009$). After the conjoint task, respondents answered an open-text question explaining their policy choice. Two human coders independently classified all responses into four categories: economic rationale, normative rationale, other rationale, and uninformative. Human inter-coder agreement was high (Cohen's $\kappa = 0.881$, 92.3% raw agreement), providing a reliable benchmark.

I classified the same responses using GPT-4o-mini and Llama 3.1 8B with a four-code codebook matching the human coding scheme. The LLMs classified English translations of the responses, matching the input the human coders used. I then compared LLM classifications to the human consensus.

The codebook required iteration. The initial version (v1) produced only fair agreement with the human coders (GPT-4o-mini $\kappa = 0.40$; Llama 3.1 8B $\kappa = 0.52$). Examining disagreements showed that both LLMs interpreted "economic" too narrowly and "uninformative" too broadly relative to the human coding. Because the conjoint is entirely about economic policy, the human coders classified policy-effectiveness language (e.g., "this policy would help them succeed") as economic, while the LLMs treated it as normative or uninformative. Broadening the economic category to include policy-effectiveness reasoning and tightening the uninformative code to cover only genuinely empty responses closed the gap across two rounds of revision.

| Comparison | Cohen's $\kappa$ | Raw agreement | Landis & Koch |
|---|---|---|---|
| Human–Human | 0.881 | 92.3% | Almost perfect |
| GPT-4o-mini–Human | 0.673 | 78.8% | Substantial |
| Llama 3.1 8B–Human | 0.611 | 74.5% | Substantial |
| GPT-4o-mini–Llama 3.1 8B | 0.717 | 81.5% | Substantial |

**Table 5:** Inter-rater agreement for the NK entrepreneurship conjoint (v3 codebook, after two rounds of codebook revision). Human–human agreement from Denney et al. (2024).

After calibration, both LLMs fall in the "substantial" agreement range ($\kappa > 0.6$), and the two LLMs agree with each other at $\kappa = 0.717$. A gap remains between LLM-human and human-human agreement, but the trajectory across codebook versions is the more important finding: two rounds of refinement moved GPT-4o-mini from $\kappa = 0.40$ (fair) to $\kappa = 0.67$ (substantial), a pattern that parallels the v1-to-v2 improvement in the main analysis. The codebook, not the model, is the primary determinant of classification quality.

This result has two implications for the main analysis. First, it provides external evidence that the iterative codebook workflow advocated in this paper produces meaningful gains in classification accuracy. Second, it calibrates expectations: even after careful iteration, LLM-human agreement ($\kappa \approx 0.6$–$0.7$) is lower than expert human-human agreement ($\kappa \approx 0.9$). The six-model cross-validation in the main analysis, which achieves Fleiss' $\kappa = 0.745$ across models of very different size and architecture, is consistent with this range and suggests that the multi-model consensus approach compensates for the gap between any single LLM and a human expert.

# Discussion and Conclusion

The results show that open-text construct validation is practical at scale. Open-weight models produce results comparable to GPT-4o-mini (Fleiss' $\kappa = 0.745$ across six classifiers), with all models agreeing on the direction and significance of treatment effects. Researchers need not depend on proprietary APIs, whose outputs may shift between model versions (Barrie et al., 2025), to conduct rigorous text classifica-

tion. The open-weight models are the reproducible core; GPT-4o-mini confirms that these alternatives match the proprietary models used in current practice.

The model size results offer practical guidance. Agreement with GPT-4o-mini increases monotonically with model size within the Qwen 2.5 family ($\kappa$: 0.744 at 3B, 0.841 at 32B, 0.860 at 72B), consistent with Chae and Davidson (2025). Yet when each model's classifications are used individually, without any majority-vote aggregation, all six produce treatment effects in the same direction for every code in every country (Supplementary Information, Table C.6). The 3B model, runnable on a consumer GPU with 8GB VRAM, finds civic commitment increasing by +9.3 pp (South Korea) and +17.7 pp (Taiwan), compared to +16.1 and +17.9 pp for GPT-4o-mini. For researchers whose primary interest is directional treatment effects on short open-text responses, even small models suffice. Mid-range models (12B–32B) offer a reasonable tradeoff between cost and classification precision for more fine-grained applications.

This approach complements existing tools for assessing respondent engagement with conjoint profiles. Prior work has validated conjoint experiments against real-world behavior (Hainmueller et al., 2015) and examined visual attention using eye-tracking (Jenke et al., 2021). Open-text responses provide a different window: how respondents *reason* about profile attributes, not just whether they attended to them. Unlike manipulation checks that condition on post-treatment responses and risk selection bias (Aronow et al., 2019), open-text validation examines the full sample. Other recent work scales open-ended responses on a continuous dimension (DiGiuseppe and Flynn, 2026); the approach here is categorical, aimed at identifying what kind of reasoning respondents use and whether the treatment shifts it.

The iterative codebook design process makes a broader point: running an unsupervised model first can catch coding mistakes that would otherwise go unnoticed. The v1 codebook grouped civic engagement with language and residence under a single code. The STM showed that these move in opposite directions under the treatment—language and residence topics *decrease* when the cue is shown—so the v2 codebook separated them. That single revision (splitting functional integration from civic commitment) changed more classification outcomes than any other design choice in the analysis.

Substantively, the consistent cross-national difference in effect size deserves attention: the cue effects are larger in Taiwan than in South Korea across both methods and all classification models. This pattern

may reflect differences in baseline expectations about immigrant civic participation, or Taiwan's more recent experience with large-scale immigration, which could make civic engagement cues more salient to Taiwanese respondents than to South Korean respondents, who have had longer exposure to immigration policy debates.

Beyond these methodological and substantive contributions, several limitations should be noted. First, the open-text responses are short (median 6–10 characters). After morphological analysis and stopword removal, 41% of South Korean and 30% of Taiwanese documents contain just a single token. STM handles this through the Bayesian prior: documents with few words receive diffuse topic assignments dominated by the prior rather than the data, which is correct inference under uncertainty but produces conservative treatment effect estimates. Filtering short documents before estimation is inadvisable because document length may itself be affected by treatment (conditioning on a post-treatment variable) and because even single words can be informative (e.g., "volunteering" unambiguously signals civic engagement). This reflects a fundamental asymmetry between the two methods: STM requires a bag-of-words representation, which means preprocessing must strip grammatical structure, particles, and conjugation before estimation can proceed. LLM classification operates on the original text, retaining the linguistic context that preprocessing necessarily removes, and draws on world knowledge to classify even very short responses confidently. The $\theta_{\max}$ sensitivity analysis (Supplementary Information, Appendix B) quantifies how short-text dilution affects STM estimates. Researchers applying STM to similarly brief text should expect attenuated effects and consider reporting such sensitivity analyses. Second, open-text responses are self-generated and may not capture everything a respondent considered; some attributes may have mattered but gone unwritten. Third, the LLM codebook reflects particular theoretical commitments, and the v1-to-v2 iteration shows that initial codebook choices meaningfully shape results. Fourth, while six-model cross-validation provides reassurance (Fleiss' $\kappa = 0.745$), individual models vary in CJK text performance. Llama 3.1 8B inflates functional integration in Taiwan (43.7% vs. 30–37% for other models), probably due to weaker multilingual capabilities. Majority-vote aggregation mitigates but does not eliminate such model-specific biases. Fifth, whether this framework generalizes beyond conjoint designs, East Asian languages, and immigration attitudes remains an open question that future work should ad-

dress. Sixth, the approach assumes that open-text responses are genuinely human-generated. Westwood et al. (2025) show that LLMs can produce survey responses realistic enough to pass standard attention checks, and self-reports suggest nontrivial rates of AI use among online respondents. If respondents generate answers with an LLM, the text no longer reflects their own reasoning, and the validation exercise breaks down. This risk reinforces the need for platform-level data quality checks alongside any open-text analysis.

These limitations notwithstanding, the core finding is clear. Combining structural topic modeling with LLM classification, and using each to inform the other, allows researchers to examine whether experimental manipulations shape respondent reasoning as intended. The primary application shows that the "cues of commitment" framework holds up in citizenship conjoint experiments across two East Asian democracies. Six independent LLM classifiers and structural topic models converge: the civic integration cue shifts respondent reasoning from practical and economic criteria toward active civic engagement. The iterative workflow, in which STM evidence exposed over-classification in the initial codebook and informed a more precise category structure, illustrates how the two methods can do more together than either does alone. The extension to an immigration admission conjoint demonstrates that this framework transfers to a different experimental design: LDA and LLM classification converge on the same constructs using a five-code scheme derived from the immigration attitudes literature, and the analysis identifies functional integration as a previously underspecified construct. The ground-truth validation against human-coded data from a third conjoint experiment confirms that iterative codebook refinement produces substantial LLM-human agreement ($\kappa \approx 0.6$–$0.7$) and that the codebook, not the model, is the primary lever for improving classification quality.

I would encourage researchers to embed open-text questions in their survey experiments, even brief ones, and to consider the two-pronged analytical approach presented here. The smallest open-weight model in this study (3B parameters) runs on a consumer GPU and produces directionally consistent results; larger models offer more precision at modest cost. The validity evidence is worth the modest additional burden of an open-text question.

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# Supplementary Information

"What Were They Thinking?" Using Open-Text Responses to Validate Constructs in Survey Experiments

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# A Survey Design and Sample

Respondents were recruited via Qualtrics using stratified quota sampling by sex, age, and geographic region to approximate national census distributions. Table A.1 presents the sample characteristics for both countries.

| | **South Korea** ($N = 1{,}999$) | **Taiwan** ($N = 2{,}050$) |
|---|---|---|
| *Gender* | | |
| Male | 1,015 (50.8%) | 1,094 (53.4%) |
| Female | 984 (49.2%) | 956 (46.6%) |
| *Age* | | |
| Mean | 46.6 | 41.3 |
| Range | 19–84 | 14–88 |
| *Education* | | |
| College or higher | 1,610 (80.6%) | 1,506 (73.5%) |
| Below college | 387 (19.4%) | 544 (26.5%) |
| *Region* | | |
| Capital area / Northern TW | 1,484 (74.3%) | 1,082 (52.8%) |
| Other regions | 513 (25.7%) | 968 (47.2%) |
| *Treatment assignment* | | |
| Cue shown | 996 (49.8%) | 1,007 (49.1%) |
| Cue not shown | 1,003 (50.2%) | 1,043 (50.9%) |

**Table A.1:** Sample characteristics. Age is in years at time of survey (2024). "Capital area" for South Korea includes Seoul, Gyeonggi, and Incheon. "Northern TW" includes Taipei, New Taipei, and Taoyuan.

Table A.2 presents the full set of 12 attributes used in the conjoint experiment on immigrant naturalization, with all attribute levels. Profiles were shown to respondents in South Korea and Taiwan; respondents chose which immigrant they considered "more deserving of citizenship." All attributes and levels were independently randomized across profiles.

| Attribute | Levels |
|---|---|
| Social integration | Only associates with fellow expatriates; Often spends time alone; Co-hosts monthly multicultural events; Volunteers weekly at a local community center |
| Ethnicity | Co-ethnic; Not co-ethnic |
| Religion | Not religious; Christian; Buddhist; Muslim |
| Country of origin | United States; China; Indonesia; Brazil (SK) / South Korea (TW); the Philippines |
| Occupation | Computer programmer; Farmer; Unemployed; Office worker |
| Family status | Single; Married; Married with children |
| Age | 27; 36; 47; 58 |
| Sex | Male; Female |
| Prior work experience | None; Worked in home country; Worked previously in country |
| Prior trips | Visited once with friends; Previously overstayed visa; Visited on short-term working visa |
| Years since arrival | 2; 7; 12 |
| Language ability | None; Broken; Conversational; Fluent |

**Table A.2:** Conjoint experiment: full attribute table (12 attributes with levels). The social integration attribute was randomly withheld from some profiles, creating the cue-shown vs. cue-not-shown treatment comparison. Country of origin levels differ slightly between the South Korea and Taiwan surveys.

# B Structural Topic Models

## B.1 Topic labels and top terms

Tables B.1 and B.2 present the topic labels and mean prevalence for each STM model. Top terms (FREX and probability) are in Korean and Traditional Chinese, respectively, and are available in the replication code.

| Topic | Label | Prevalence (%) |
|---|---|---|
| 1 | Community Engagement | 18.8 |
| 2 | Language & Visa | 11.9 |
| 3 | Religious/Ethnic Preference | 22.9 |
| 4 | Residence & Assimilation | 21.3 |
| 5 | Experience & Profession | 12.1 |
| 6 | Employment & Family | 13.1 |

**Table B.1:** South Korea STM topics ($K = 6$). Prevalence is the mean document-topic proportion across the corpus.

| Topic | Label | Prevalence (%) |
|---|---|---|
| 1 | Fluency & Impression | 18.6 |
| 2 | Life & Religion | 6.2 |
| 3 | Work & Integration | 18.5 |
| 4 | Residence & Occupation | 9.7 |
| 5 | Understanding Taiwan | 11.4 |
| 6 | Work Skills & Language | 8.1 |
| 7 | Community Participation & Identity | 14.6 |
| 8 | Residence Record & Overstay | 12.8 |

**Table B.2:** Taiwan STM topics ($K = 8$). Prevalence is the mean document-topic proportion across the corpus.

## B.2 Topic assignment clarity and threshold sensitivity

Structural topic models assign every document a probability distribution over all $K$ topics, regardless of content quality. A substantive response that clearly discusses a single theme receives a concentrated distribution (high $\theta_{\max}$), while an uninformative or ambiguous response receives a diffuse distribution

spread across topics. This section examines the clarity of STM topic assignments and how treatment effect estimates change when diffusely assigned documents are excluded.

The open-text responses in this study are extremely short. After morphological analysis (kiwipiepy for Korean, quanteda for Chinese) and stopword removal, 41% of South Korean documents and 30% of Taiwanese documents contain a single token; the median is 2 tokens in both countries. Standard STM preprocessing drops documents that produce zero features (265 of 1,999 in South Korea, 145 of 2,050 in Taiwan), but no minimum word count threshold is applied, consistent with standard practice in the STM literature (Roberts et al., 2014). STM handles short documents through its Bayesian prior: documents with few words receive posterior topic proportions dominated by the prior rather than the data, producing appropriately diffuse assignments that express the model's uncertainty.

Figure B.1 shows the distribution of $\theta_{\text{max}}$ (the maximum topic probability for each document) in both countries. The dashed orange line marks the uniform distribution ($1/K$: 0.167 for South Korea, 0.125 for Taiwan), and the dotted blue line marks a moderate threshold of $\theta_{\text{max}} = 0.30$. In South Korea ($K = 6$), the median $\theta_{\text{max}}$ is 0.315, and only 3.2% of documents exceed 0.50. In Taiwan ($K = 8$), the median is 0.303 with 11.5% above 0.50. The diffuse assignments are a direct consequence of the short document lengths: with 1–2 tokens, the posterior cannot move far from the prior.

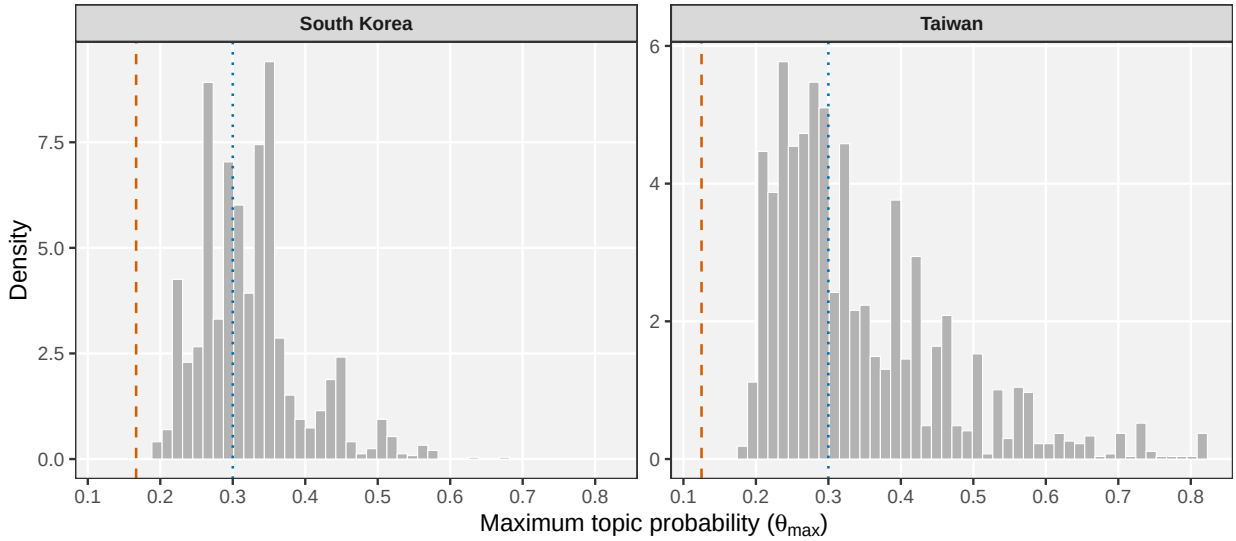


**Figure B.1:** Distribution of maximum topic probability ($\theta_{\text{max}}$) for each document. Dashed orange: uniform baseline ($1/K$). Dotted blue: moderate filtering threshold ($\theta_{\text{max}} = 0.30$). Most documents are diffusely assigned across topics, especially in South Korea where $K = 6$ constrains the topic space.

Figure B.2 shows how the civic topic treatment effect changes as the minimum $\theta_{\text{max}}$ threshold in-

creases (i.e., as diffusely assigned documents are progressively excluded). The solid lines show STM civic topic effects at each threshold; the dashed horizontal lines show the LLM majority-vote civic commitment estimates for comparison.

As the threshold rises, the STM civic effect increases monotonically in both countries. At $\theta_{\max} \geq 0.30$ (retaining 59% of South Korean and 51% of Taiwanese documents), the effects rise to +10.5 pp and +18.2 pp, respectively. The STM estimates cross the LLM estimates around $\theta_{\max} \approx 0.35$–$0.40$, and at higher thresholds the STM effects substantially exceed the LLM estimates (reflecting the smaller and potentially unrepresentative samples that remain).

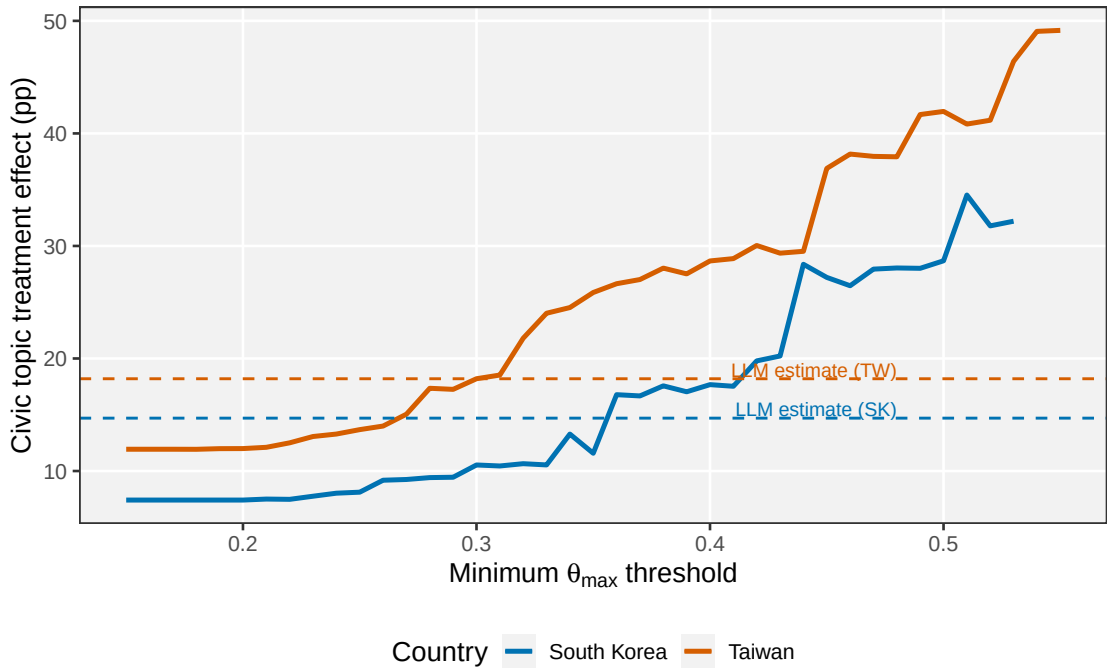


**Figure B.2:** Sensitivity of the civic topic treatment effect to $\theta_{\max}$ thresholds. Solid lines: STM civic topic effect (mean $\theta$ difference, cue-shown minus cue-not-shown). Dashed lines: LLM majority-vote civic commitment estimates. As diffusely assigned documents are excluded, the STM effect increases and approaches or exceeds the LLM estimate.

Table B.3 summarizes these results at selected thresholds.

| **Min.** $\theta_{\max}$ | **South Korea** | | **Taiwan** | |
|---|---|---|---|---|
| | Docs retained | Civic effect (pp) | Docs retained | Civic effect (pp) |
| 0 (all) | 1,734 (100%) | +7.4 | 1,905 (100%) | +11.9 |
| 0.25 | 1,499 (86%) | +8.1 | 1,463 (77%) | +13.7 |
| 0.30 | 1,014 (59%) | +10.5 | 964 (51%) | +18.2 |
| 0.35 | 505 (29%) | +11.6 | 681 (36%) | +25.9 |
| 0.40 | 226 (13%) | +17.7 | 476 (25%) | +28.7 |
| *LLM majority-vote (for comparison)* | | | | |
| — | 1,820 (91%) | +14.7 | 1,938 (94%) | +18.2 |

**Table B.3:** Civic topic treatment effect at different $\theta_{\max}$ thresholds. The STM effect increases as diffusely assigned documents are excluded. At $\theta_{\max} \geq 0.30$, STM estimates approach the LLM majority-vote estimates. The LLM row shows the majority-vote civic commitment effect for comparison; the "docs retained" count for the LLM excludes responses without majority agreement (7.2% of the corpus).

This analysis illustrates a structural difference between the two methods. STM distributes every document, including uninformative ones, across the entire topic space. This dilutes treatment effects because noise documents receive non-trivial proportions on treatment-sensitive topics. LLM classification handles this differently: the explicit `other` category absorbs uninformative responses, effectively filtering them before treatment effects are computed on the substantive codes. The $\theta_{\max}$ threshold analysis is the STM analogue of this filtering. As the threshold increases and noisy documents are removed, the STM estimates converge toward (and eventually exceed) the LLM estimates, confirming that the difference in effect magnitudes between the two methods is driven by how each handles uninformative text rather than by substantive disagreement about the treatment effect.

# C LLM Classification

## C.1 Codebook

The codebook follows the structured format recommended by Halterman & Keith (2025), with each code defined by a label, definition, clarification (inclusion criteria), and negative clarification (exclusion criteria). Each code also includes few-shot examples in both Korean and Traditional Chinese.

| Code | Definition and Scope |
|---|---|
| `civic_commitment` | Participation in community life: volunteering, community organizing, social participation, cultural events, helping neighbors. The respondent cites engagement directed outward toward the community. |
| `functional_integration` | Individual-level adaptation: language competence, residence duration, law-abiding behavior. The respondent cites practical prerequisites for functioning in the host country, rather than participation in community life. |
| `identity_concern` | Ethnic, cultural, or national identity concerns. Includes religion and ethnicity when used as criteria for evaluating immigrants. |
| `economic_focus` | Economic contributions and qualifications: jobs, taxes, career, occupation, work experience, financial self-sufficiency. |
| `other` | Truly uninformative responses: filler words, gibberish, meta-comments about the survey. Brevity alone is never sufficient reason for this code. |

**Table C.1:** LLM classification codebook (v2, 5 codes). The main refinement from v1 was separating `functional_integration` from `civic_commitment` based on STM evidence that language and residence topics decrease with the civic integration cue.

## C.2 Infrastructure and parse errors

Table C.2 provides technical details for each classifier. All models used temperature $= 0$, seed $= 42$, and a maximum of 200 output tokens. Open-weight models were served via Ollama.

| Model | Parameters | Infrastructure | Parse Errors |
|---|---|---|---|
| GPT-4o-mini | — | OpenAI API | 0 |
| Qwen 2.5 3B | 3B | Consumer GPU (NVIDIA RTX 3060 Ti, 8GB) | 38 |
| Llama 3.1 8B | 8B | Consumer GPU (NVIDIA RTX 3060 Ti, 8GB) | 0 |
| Gemma 3 12B | 12B | University HPC, NVIDIA A100 | 1 |
| Qwen 2.5 32B | 32B | University HPC, NVIDIA A100, 4-bit quantization | 0 |
| Qwen 2.5 72B | 72B | University HPC, NVIDIA A100 80GB, 4-bit quantization | 0 |

**Table C.2:** LLM infrastructure and parse error counts. Parse errors indicate responses where the model output could not be parsed into a valid JSON classification; these responses are excluded from agreement and prevalence calculations. Qwen 2.5 3B accounts for nearly all parse errors (38 of 39 total across all models), consistent with smaller models producing less reliable structured output.

## C.3 Per-model code distributions

Table C.3 presents the code distribution for each of the six LLM classifiers, by country.

| Model | Country | Civic | Functional | Identity | Economic | Other |
|---|---|---|---|---|---|---|
| GPT-4o-mini | South Korea | 11.7 | 28.5 | 21.2 | 17.5 | 21.0 |
| GPT-4o-mini | Taiwan | 10.8 | 31.1 | 14.6 | 21.1 | 22.3 |
| Qwen 2.5 3B | South Korea | 10.7 | 19.5 | 20.4 | 21.4 | 28.1 |
| Qwen 2.5 3B | Taiwan | 11.3 | 21.2 | 16.8 | 25.2 | 25.5 |
| Llama 3.1 8B | South Korea | 14.7 | 36.4 | 21.1 | 18.5 | 9.4 |
| Llama 3.1 8B | Taiwan | 10.7 | 43.7 | 16.2 | 14.8 | 14.6 |
| Gemma 3 12B | South Korea | 12.5 | 35.5 | 22.3 | 16.9 | 12.7 |
| Gemma 3 12B | Taiwan | 10.9 | 36.7 | 15.0 | 22.0 | 15.4 |
| Qwen 2.5 32B | South Korea | 7.5 | 33.7 | 20.0 | 18.7 | 20.1 |
| Qwen 2.5 32B | Taiwan | 9.0 | 32.1 | 15.6 | 22.8 | 20.5 |
| Qwen 2.5 72B | South Korea | 9.4 | 30.7 | 18.9 | 19.4 | 21.6 |
| Qwen 2.5 72B | Taiwan | 10.7 | 30.0 | 14.7 | 23.4 | 21.2 |

**Table C.3:** Code distribution by model and country (% of classified responses). Llama 3.1 8B shows elevated functional integration in Taiwan (43.7%) relative to other models (28.5–36.7%), likely reflecting weaker CJK language performance. Qwen 2.5 3B shows elevated `other` rates (25–28%), consistent with lower classification confidence at smaller model sizes.

## C.4 Cross-model agreement

Table C.4 presents pairwise raw agreement rates and Cohen's $\kappa$ for all 15 model pairs.

| Model 1 | Model 2 | Agreement (%) | Cohen's $\kappa$ |
|---|---|---|---|
| GPT-4o-mini | Qwen 2.5 3B | 79.7 | 0.744 |
| GPT-4o-mini | Llama 3.1 8B | 77.2 | 0.706 |
| GPT-4o-mini | Gemma 3 12B | 83.9 | 0.792 |
| GPT-4o-mini | Qwen 2.5 32B | 87.6 | 0.841 |
| GPT-4o-mini | Qwen 2.5 72B | 89.1 | 0.860 |
| Qwen 2.5 3B | Llama 3.1 8B | 66.3 | 0.578 |
| Qwen 2.5 3B | Gemma 3 12B | 71.8 | 0.646 |
| Qwen 2.5 3B | Qwen 2.5 32B | 77.2 | 0.710 |
| Qwen 2.5 3B | Qwen 2.5 72B | 79.6 | 0.741 |
| Llama 3.1 8B | Gemma 3 12B | 79.0 | 0.722 |
| Llama 3.1 8B | Qwen 2.5 32B | 75.7 | 0.683 |
| Llama 3.1 8B | Qwen 2.5 72B | 76.8 | 0.700 |
| Gemma 3 12B | Qwen 2.5 32B | 85.7 | 0.815 |
| Gemma 3 12B | Qwen 2.5 72B | 85.8 | 0.816 |
| Qwen 2.5 32B | Qwen 2.5 72B | 91.2 | 0.886 |
| *Fleiss' multi-rater $\kappa$* | | | **0.745** |

**Table C.4:** Pairwise agreement rates and Cohen's $\kappa$ for all 15 model pairs ($N = 4{,}049$ responses). Agreement increases with model size: within the Qwen 2.5 family, $\kappa$ with GPT-4o-mini rises from 0.744 (3B) to 0.841 (32B) to 0.860 (72B). The highest pairwise agreement is between Qwen 2.5 32B and 72B ($\kappa = 0.886$).

## C.5 Model confidence

Each LLM classifier reports a self-assessed confidence level (HIGH, MEDIUM, or LOW) alongside its classification. Table C.5 summarizes the proportion of HIGH-confidence classifications by model and country.

| Model | South Korea (% HIGH) | Taiwan (% HIGH) |
|---|---|---|
| GPT-4o-mini | 80.9 | 87.9 |
| Qwen 2.5 3B | 69.0 | 69.9 |
| Llama 3.1 8B | 49.3 | 60.8 |
| Gemma 3 12B | 74.9 | 81.6 |
| Qwen 2.5 32B | 53.3 | 70.0 |
| Qwen 2.5 72B | 48.3 | 60.0 |

**Table C.5:** Proportion of HIGH-confidence classifications by model and country. GPT-4o-mini reports the highest confidence. Larger open-weight models do not uniformly report higher confidence than smaller ones, indicating that self-reported confidence reflects model-specific calibration rather than classification quality.

## C.6 Per-model cue effects

Table C.6 presents the treatment effect of the social integration cue on each classification code for every model individually, without majority-vote aggregation. Each row shows the change in code prevalence (in percentage points) when the cue is shown versus not shown, computed from that model's full set of classifications.

All six models, from the smallest (Qwen 2.5 3B, runnable on a consumer GPU) to the largest (Qwen 2.5 72B, requiring a university HPC node), agree on the direction of every substantive effect. Civic commitment increases in both countries across all models; functional integration and economic focus decrease; identity concern is stable.

| Model | Country | Civic Commitment | Functional Integration | Economic Focus | Identity Concern |
|---|---|---|---|---|---|
| *Proprietary* | | | | | |
| GPT-4o-mini | South Korea | +16.1 | −7.8 | −5.7 | −4.7 |
| GPT-4o-mini | Taiwan | +17.9 | −8.1 | −10.1 | −1.2 |
| *Open-weight (Qwen 2.5 family, size scaling)* | | | | | |
| Qwen 2.5 3B | South Korea | +9.3 | −3.0 | −6.7 | −1.4 |
| Qwen 2.5 3B | Taiwan | +17.7 | −6.8 | −13.2 | +1.8 |
| Qwen 2.5 32B | South Korea | +13.1 | −5.8 | −4.6 | −4.3 |
| Qwen 2.5 32B | Taiwan | +16.5 | −4.0 | −10.9 | −1.7 |
| Qwen 2.5 72B | South Korea | +14.3 | −7.0 | −6.1 | −3.0 |
| Qwen 2.5 72B | Taiwan | +18.1 | −6.8 | −11.9 | −0.6 |
| *Open-weight (other model families)* | | | | | |
| Llama 3.1 8B | South Korea | +12.2 | −7.5 | −4.4 | −2.6 |
| Llama 3.1 8B | Taiwan | +15.9 | −10.9 | −6.5 | −0.9 |
| Gemma 3 12B | South Korea | +14.3 | −6.0 | −5.5 | −4.6 |
| Gemma 3 12B | Taiwan | +14.4 | −4.3 | −10.4 | −0.7 |

**Table C.6:** Change in code prevalence (percentage points) when the social integration cue is shown, by model and country. Each model's classifications are used independently; no majority-vote aggregation is applied. The `other` category is omitted. All models agree on the direction of all substantive effects. The Qwen 2.5 family (3B, 32B, 72B) permits direct comparison of model size effects within the same architecture.

## C.7 Cue effect by code (majority vote)

Table C.7 presents the prevalence of each substantive code when the social integration cue was shown versus not shown, using majority-vote classification (4+ of 6 models agree). The `other` category is excluded.

| Country | Code | No Cue (%) | Cue (%) | Diff (pp) |
|---|---|---|---|---|
| South Korea | Civic Commitment | 2.4 | 20.9 | +18.5 |
| South Korea | Functional Integration | 44.2 | 35.0 | −9.2 |
| South Korea | Identity Concern | 27.1 | 24.3 | −2.8 |
| South Korea | Economic Focus | 26.3 | 19.8 | −6.5 |
| Taiwan | Civic Commitment | 1.3 | 24.4 | +23.1 |
| Taiwan | Functional Integration | 43.7 | 35.6 | −8.1 |
| Taiwan | Identity Concern | 19.7 | 18.7 | −1.0 |
| Taiwan | Economic Focus | 35.3 | 21.3 | −14.0 |

**Table C.7:** Prevalence of each classification code by treatment condition, using majority-vote classification (4+ of 6 models agree). Differences are in percentage points. The `other` category is excluded; percentages are computed among substantive codes only.